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MAAE 2300: Fluid Mechanics I

Tutorial Session 1: Fluid Properties & Dimensional Homogeneity

Session 1: Agenda & Problem Breakdown

☰ Problem Set 1 Roadmap

- 1 **Problem 1:** Bulk Modulus & Ocean Water Compression (10 km).
- 2 **Problem 2:** Helium Blimp & Ideal Gas Law (Imperial Units).
- 3 **Problem 3:** Aircraft Oxygen Tank & Cabin Pressure (ISA Atmosphere).
- 4 **Problem 4:** Dimensional Homogeneity (Stokes-Oseen & Viscometer).
- 5 **Problem 5:** Viscous Resistance on a Sliding Piston.
- 6 **Problem 6:** Concentric-Cylinder Viscometer (Cylinder vs. Bottom Gap).
- 7 **Problem 7:** Water Strider Supported by Surface Tension.
- 8 **Problem 8:** Soap Bubble Coalescence (Isothermal Gas Expansion).

🎯 Key Learning Objectives

- Distinguish between absolute and gauge pressure systematically.
- Apply the continuum hypothesis and fluid properties (ρ, μ, E_v, γ).
- Use Newton's law of viscosity in shear-driven geometries.
- Formulate surface tension balances on contact lines and bubbles.
- Verify dimensional homogeneity in empirical and analytical equations.

Quick-Reference

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Fluid Properties: Governing Relations Quick-Card

Thermodynamic & Elastic Properties

Ideal Gas Law: $P = \rho RT \implies \rho = \frac{P}{RT}$ (P, T absolute!)

Specific Gravity: $SG = \frac{\rho}{\rho_{H_2O, ref}}$ ($\rho_{ref} = 1000 \text{ kg m}^{-3}$)

Bulk Modulus of Elasticity (K or E_v):

$$K = -\frac{\Delta P}{\Delta V/V} \implies \Delta V = -\frac{\Delta P \cdot V}{K}$$

 Absolute vs. Gauge Pressure

$$P_{abs} = P_{gauge} + P_{atm} \quad P_{abs} = P_{atm} - P_{vac}$$

Viscosity & Surface Tension

Newton's Law of Viscosity (1D): $\tau = \mu \frac{du}{dy}$

Kinematic Viscosity: $\nu = \frac{\mu}{\rho}$ [$\text{m}^2 \text{s}^{-1}$] (1 cSt = $10^{-6} \text{ m}^2 \text{s}^{-1}$)

Young-Laplace Pressure Jump across Curved Interface:

- Liquid droplet (1 interface): $\Delta P = \frac{2\gamma}{R} = \frac{4\gamma}{D}$
- Soap bubble (2 interfaces): $\Delta P = \frac{4\gamma}{R} = \frac{8\gamma}{D}$

Surface Tension Force Along Contact Line

$$F_\sigma = \gamma L_{\text{interface}} \cos \theta \quad (\theta = \text{contact angle})$$

Problem 1

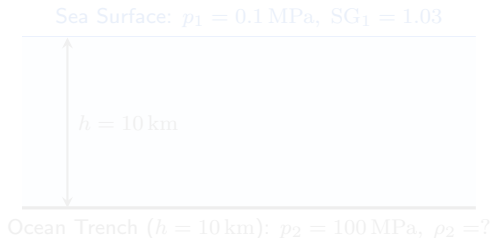
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Problem 1

The pressure of the water at a depth of 10 km in the ocean is about 100 MPa (sea level atmospheric pressure is about 0.1 MPa). Assuming a value of the bulk modulus of elasticity $K = 2100$ MPa (E_v), what is the density of the water at that depth, if the S.G. (specific gravity) is 1.03 at the surface?

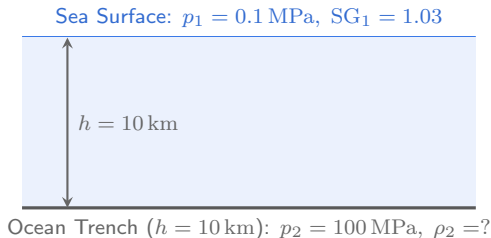
(Ans: 1081 kg m^{-3})



Problem 1

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(Ans: 1081 kg m^{-3})



Problem 1: Solution & Bulk Modulus Analysis

Engineering Assumptions

- Bulk modulus of water $K = 2100 \text{ MPa}$ is constant over ΔP .
- Closed control mass: mass m is conserved during compression.
- Surface density:
 $\rho_1 = 1.03 \times 1000 = 1030 \text{ kg m}^{-3}$.

Bulk Modulus Definition (K)

$$K = -\frac{\Delta P}{\Delta V/V} \Rightarrow \Delta V = -\frac{\Delta P \cdot V}{K}$$

Conservation of mass: $m = \rho_1 V_1 = \rho_2 V_2 = \text{const.}$

➔ **Given:** $P_1 = 0.1 \text{ MPa}$, $P_2 = 100 \text{ MPa}$,
 $\Delta P = 99.9 \text{ MPa}$, $K = 2100 \text{ MPa}$.

🎯 **Find:** Water density at oceanic depth ρ_2 .

➡ 1. Reference Volume Method ($V_1 = 1 \text{ m}^3$)

Surface parcel mass: $m = \rho_1 V_1 = (1030)(1) = 1030 \text{ kg}$.

Volume deformation under $\Delta P = 99.9 \text{ MPa}$:

$$\Delta V = -\frac{\Delta P V_1}{K} = -\frac{99.9 \times 1}{2100} = -0.04757 \text{ m}^3$$

$$V_2 = V_1 + \Delta V = 1 - 0.04757 = 0.95243 \text{ m}^3$$

✔ 2. Density at Oceanic Depth

Conserved parcel mass $m = 1030 \text{ kg}$:

$$\begin{aligned}\rho_2 &= \frac{m}{V_2} = \frac{1030 \text{ kg}}{0.95243 \text{ m}^3} \\ &= 1081.4 \text{ kg m}^{-3} \approx 1081 \text{ kg m}^{-3}\end{aligned}$$

Direct formula: $\rho_2 = \frac{\rho_1}{1 - \Delta P/K} = 1081 \text{ kg m}^{-3}$
(+4.87%).

Problem 2

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The helium-filled blimp below is used at various athletic events. Determine the number of pounds of helium within it if its volume is $68,000 \text{ ft}^3$ and the temperature and pressure are 80°F and 14.2 psi , respectively.

(Ans: 668 lb)



Figure: Blimp sketch.

Problem 2: Solution & Ideal Gas Analysis

Assumptions

- Helium behaves as an ideal gas at atmospheric conditions.
- Uniform temperature and pressure throughout the envelope.

Ideal Gas Law

$$P = \rho RT \implies m = \frac{PV}{RT}$$

Critical rule: P and T must be expressed in absolute units!

➔ **Given:** $V = 68,000 \text{ ft}^3$, $T = 80^\circ\text{F}$, $P = 14.2 \text{ psi}$.

🎯 **Find:** Total weight/mass of helium m (lb).

➔ 1. Absolute Units & Ideal Gas Evaluation

- $T = 80 + 459.67 \approx 540^\circ\text{R}$,
 $P = 14.2 \times 144 = 2044.8 \text{ lbf/ft}^2$
- $R_{\text{He}} = \frac{R_u}{M} = \frac{1545.3}{4.003} = 386.0 \text{ ft} \cdot \text{lbf}/(\text{lbm} \cdot ^\circ\text{R})$
 $m = \frac{PV}{RT} = \frac{(2044.8)(68,000)}{(386.0)(540)} = \mathbf{668 \text{ lb}}$

Physical Takeaway

Air displaced at standard conditions weighs $\sim 5,100 \text{ lb}$, providing a net buoyant lifting force of $\approx 4,400 \text{ lb}$ for the gondola and payload.

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An aircraft carries an oxygen cylinder to which a pressure gauge is attached. The gauge reads 100 kPa(g). The air inside the aircraft is pressurized to 8 psi above the ambient pressure outside the aircraft, which is flying at 12,000 m in a standard atmosphere (ISA Table provided on the last page). What is the absolute pressure of the cabin air and of the oxygen in the tank?

(Ans: 74.56 kPa, 174.56 kPa)

Remarks:

- Gauge pressure is measured relative to local atmospheric/cabin pressure; absolute pressure is measured relative to absolute vacuum.
- From Standard Atmosphere table: $p_{\text{atm}} = 19.40 \text{ kPa}$ at $z = 12,000 \text{ m}$.

Problem 3: Solution & Absolute vs. Gauge Pressure

Pressure Definitions

$$P_{\text{abs}} = P_{\text{ambient}} + P_{\text{gauge}}$$

- Ambient pressure outside:
 $p_{\text{outside}} = 19.40 \text{ kPa}$.
- Cabin pressure increment: $\Delta p = 8 \text{ psi}$.
- Cylinder gauge is located **inside the cabin**.

➔ 1. Conversion to SI Units

$$\Delta p_{\text{cabin}} = 8 \text{ psi} \times \frac{6.8948 \text{ kPa}}{1 \text{ psi}} = 55.16 \text{ kPa}$$

➔ 2. Cabin Absolute Pressure

$$\begin{aligned} p_{\text{cabin,abs}} &= p_{\text{outside}} + \Delta p_{\text{cabin}} \\ &= 19.40 + 55.16 = 74.56 \text{ kPa} \end{aligned}$$

Equivalent altitude: $\sim 2400 \text{ m}$ cabin pressurization altitude.

➔ 3. Oxygen Tank Absolute Pressure

The tank gauge measures relative to the cabin environment:

$$\begin{aligned} p_{\text{tank,abs}} &= p_{\text{cabin,abs}} + p_{\text{gauge,tank}} \\ &= 74.56 + 100 = 174.56 \text{ kPa} \end{aligned}$$

Final Answers: $p_{\text{cabin}} = 74.56 \text{ kPa}$, $p_{\text{tank}} = 174.56 \text{ kPa}$.

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Problem 4: Dimensional Homogeneity

4.1 Stokes–Oseen Formula

According to Stokes–Oseen formula, the drag force F on a sphere of diameter D in a fluid stream of low velocity V , density ρ , and viscosity μ is given by:

$$F = 3\pi \mu D V + \frac{9\pi}{16} \rho V^2 D^2$$

Is the formula dimensionally consistent? If yes, prove it. If no, what should be the dimensions of the numerical constants to make it consistent?

(Ans: Yes)

Problem 4.1: Dimensional Consistency Proof

➔ 1. Dimensions of Variables (M, L, T)

- Force: $[F] = MLT^{-2}$
- Viscosity: $[\mu] = ML^{-1}T^{-1}$
- Velocity: $[V] = LT^{-1}$
- Diameter: $[D] = L$
- Density: $[\rho] = ML^{-3}$
- Dimensionless constants: $[3\pi] = [9\pi/16] = 1$

➔ 2. Dimensional Check of Terms

Term 1 (Viscous Stokes Drag):

$$\begin{aligned}[3\pi\mu DV] &= (ML^{-1}T^{-1})(L)(LT^{-1}) \\ &= \mathbf{MLT^{-2}} \quad (\text{Force})\end{aligned}$$

Term 2 (Inertial Oseen Correction):

$$\begin{aligned}\left[\frac{9\pi}{16}\rho V^2 D^2\right] &= (ML^{-3})(L^2T^{-2})(L^2) \\ &= \mathbf{MLT^{-2}} \quad (\text{Force})\end{aligned}$$

✓ Conclusion 4.1

Both additive terms have identical dimensions of force $[MLT^{-2}]$. **Yes, the formula is dimensionally consistent!**

4.2 Aerodynamic Pitching Moment

The pitching aerodynamics moment M_o on an aircraft wing is a function of airspeed V , wing area A , chord C , and density ρ such that $M_o = \beta V^2 A C \rho$. What are the dimensions of the coefficient β ? **(Ans: Dimensionless)**

4.3 Falling Sphere Viscometer

Suggested formulas to infer viscosity μ and fall time t for a sphere of net weight W_{net} , diameter D , falling distance L :

$$\mu \approx \frac{W_{\text{net}} t}{3\pi D L}, \quad t \geq \frac{2\rho D L}{\mu}$$

Are the above formulas dimensionally homogeneous? **(Ans: Yes)**

Problem 4.2 & 4.3: Solutions

➡ 4.2 Pitching Moment Coefficient

$$[M_o] = \text{Force} \times \text{Length} = ML^2T^{-2}.$$

Evaluating right-hand side group:

$$\begin{aligned}[V^2AC\rho] &= (L^2T^{-2})(L^2)(L)(ML^{-3}) \\ &= ML^2T^{-2}\end{aligned}$$

Equating dimensions:

$$[\beta] = \frac{[M_o]}{[V^2AC\rho]} = \frac{ML^2T^{-2}}{ML^2T^{-2}} = 1 \quad (\text{Dimensionless})$$

➡ 4.3 Viscometer Equations Check

Formula 1 (μ):

$$\left[\frac{W_{\text{net}} t}{DL} \right] = \frac{(MLT^{-2})T}{L^2} = ML^{-1}T^{-1} = [\mu]$$

Formula 2 (t):

$$\left[\frac{\rho DL}{\mu} \right] = \frac{(ML^{-3})L^2}{ML^{-1}T^{-1}} = T = [t]$$

✓ Conclusion 4.3

Both viscometer relations are **dimensionally homogeneous!**

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Problem 5

A smooth-sided piston 5 cm long and of diameter 5.000 cm moves within a cylinder of 5.010 cm diameter. The lubricating oil has a viscosity of $2 \times 10^{-3} \text{ N} \cdot \text{s}/\text{m}^2$. What force is required to move the piston at a speed of 12 m/s? Assume the velocity in the gap between the cylinder and the piston varies linearly, as indicated in the sketch.

(Ans: 3.77 N)

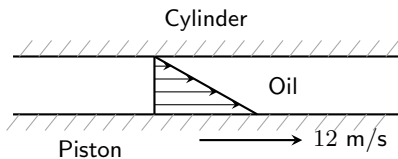


Figure: Sketch.

Problem 5: Solution & Viscous Shear Force

⚙ Assumptions

- No-slip condition at both solid surfaces.
- Linear velocity profile across the narrow radial gap ($h \ll D$).
- Newtonian lubricating oil ($\mu = \text{const}$).

⚖ Newton's Law of Viscosity

$$\tau = \mu \frac{du}{dy} = \mu \frac{U}{h}$$

Viscous resisting shear force: $F = \tau A$.

➡ 1. Clearance Geometry & Surface Area

Radial clearance gap h :

$$h = \frac{D_c - D}{2} = \frac{5.010 - 5.000}{2} \text{ cm} = 5 \times 10^{-5} \text{ m}$$

Wetted cylindrical contact area A :

$$A = \pi D L = \pi (0.05 \text{ m}) (0.05 \text{ m}) = 7.854 \times 10^{-3} \text{ m}^2$$

✔ Required Driving Force

$$\begin{aligned} F &= \tau A = \left(\mu \frac{U}{h} \right) (\pi D L) \\ &= \left(2 \times 10^{-3} \frac{12}{5 \times 10^{-5}} \right) (7.854 \times 10^{-3}) = 3.77 \text{ N} \end{aligned}$$

Problem 6

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Problem 6

The drawing shows a concentric-cylinder viscometer. The fluid fills the gap between the two cylinders. The falling weight unreels a string from the pulley, thus rotating the inner cylinder. The viscosity of the fluid is then inferred from the rotational speed of the cylinder. Assume that the fluid velocity varies linearly both in the vertical gap between the cylinders and in the gap between the flat bottom of the inner cylinder and the bottom of the outer cylinder. The viscometer has the following dimensions: $R = 6$ cm, $R_{\text{pulley}} = 3$ cm, $L = 15$ cm, and $t = h = 0.1$ cm. When the falling weight has a mass of 30 g, the cylinder is observed to rotate at 30 RPM.

6.1.

Estimate the viscosity of the fluid neglecting the frictional torque due to the fluid in the bottom gap.

(Ans: $1.3805 \times 10^{-2} \text{ N} \cdot \text{s/m}^2$)

Problem 6.1: Viscosity Neglecting Bottom Gap

➔ 1. Drive Torque & Angular Speed

$$\omega = \frac{30 \times 2\pi}{60} = \pi \approx 3.1416 \text{ rad s}^{-1}$$

$$\begin{aligned} T_{\text{drive}} &= mgR_{\text{pulley}} \\ &= (0.030 \text{ kg})(9.81 \text{ m s}^{-2})(0.03 \text{ m}) \\ &= 8.829 \times 10^{-3} \text{ N m} \end{aligned}$$

➔ 2. Cylindrical Gap Torque Balance

At $r = R$, linear wall velocity is $U = \omega R$:

$$\tau = \mu \frac{\omega R}{t}$$

$$T_{\text{cyl}} = \tau AR = \left(\mu \frac{\omega R}{t} \right) (2\pi RL) R = \frac{2\pi\mu\omega R^3 L}{t}$$

✓ 6.1 Inferred Viscosity

Equating $T_{\text{drive}} = T_{\text{cyl}}$:

$$\mu = \frac{T_{\text{drive}} t}{2\pi\omega R^3 L} = 1.381 \times 10^{-2} \text{ N s m}^{-2}$$

6.2.

Repeat (6.1) but now include the effect of the fluid in the bottom gap.

(Ans: $1.255 \times 10^{-2} \text{ N} \cdot \text{s}/\text{m}^2$)

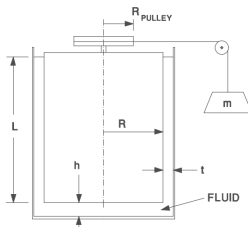


Figure: Concentric-cylinder viscometer.

Problem 6.2: Solution Including Bottom Disk Shear

➔ 3. Bottom Disk Gap Torque Integration

At radius $r \in [0, R]$: speed $u(r) = \omega r$, shear $\tau(r) = \mu \frac{\omega r}{h}$.

Differential torque on ring $dA = 2\pi r dr$:

$$dT_{\text{bottom}} = r \tau(r) dA = \frac{2\pi\mu\omega}{h} r^3 dr$$

Integrating over circular bottom:

$$T_{\text{bottom}} = \frac{2\pi\mu\omega}{h} \int_0^R r^3 dr = \frac{\pi\mu\omega R^4}{2h}$$

➔ 4. Total Torque Balance & Inferred Viscosity

Torque balance: $T_{\text{drive}} = \mu\omega\pi R^3 \left(\frac{2L+R/2}{t} \right)$.

Solving for dynamic viscosity μ ($t = h = 1 \times 10^{-3} \text{ m}$):

$$\begin{aligned} \mu &= \frac{T_{\text{drive}}}{\omega\pi R^3 \left(\frac{2(0.15)+0.03}{10^{-3}} \right)} \\ &= 1.255 \times 10^{-2} \text{ N s m}^{-2} \end{aligned}$$

💡 Pedagogical Takeaway

Neglecting bottom gap overestimated μ by +10%.
The end effect in viscometers is non-negligible!

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Problem 7

The water strider insect below is supported on the water surface by surface tension. Assuming a saline water with density ρ and surface tension γ , insect weight W that is equally distributed over four legs only, and a water–insect leg interface tangent of angle θ measured from the vertical, derive an expression for the minimum length of the water–insect leg interface, $L_{\min}$, to support its weight on the water surface.

(Ans: $W/(8\gamma \cos \theta)$)

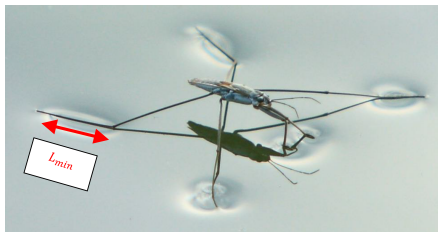


Figure: Water strider.

Problem 7: Solution & Surface Tension Equilibrium

➔ 1. Load per Leg

Since the insect weight W is carried equally by 4 legs:

$$W_{\text{leg}} = \frac{W}{4}$$

➔ 2. Surface Tension Interface Lines

Each leg produces a depression forming **two contact lines** with the water surface (left and right of the leg).

$$L_{\text{contact,total}} = 2 \times L_{\text{min}}$$

Upward vertical surface tension component:

$$F_{\sigma,y} = 2 \times (\gamma L_{\text{min}}) \times \cos \theta$$

➔ 3. Vertical Force Balance & Min Length

For static equilibrium of a single leg ($\sum F_y = 0$):

$$W_{\text{leg}} = F_{\sigma,y} \implies \frac{W}{4} = 2\gamma L_{\text{min}} \cos \theta$$

Solving for minimum interface contact length L_{min} :

$$L_{\text{min}} = \frac{W}{8\gamma \cos \theta}$$

💡 Physical Insight

If the insect leg is non-wetting / superhydrophobic ($\theta \rightarrow 0^\circ$), $\cos \theta = 1$, minimizing the required contact length: $L_{\text{min}} = W/(8\gamma)$.

Problem 8

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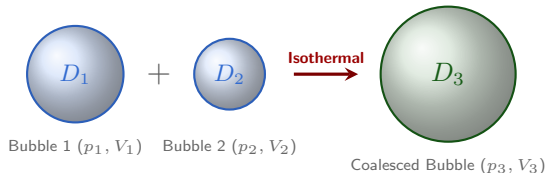
A soap bubble of diameter D_1 coalesces with another bubble of diameter D_2 to form a single bubble D_3 with the same amount of air. For an isothermal process (constant temperature), express D_3 as a function of D_1 , D_2 , p_{atm} , and surface tension γ .

(Ans: $p_{\text{atm}}D_3^3 + 8\gamma D_3^2 = (p_{\text{atm}}D_1^3 + 8\gamma D_1^2) + (p_{\text{atm}}D_2^3 + 8\gamma D_2^2)$)



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Problem 8: Solution & Isothermal Coalescence

Young-Laplace for Soap Bubbles

A soap bubble consists of a thin liquid film with **two interfaces** (inner and outer):

$$\Delta P = P_{\text{inside}} - p_{\text{atm}} = \frac{4\gamma}{R} = \frac{8\gamma}{D}$$

$$P_i = p_{\text{atm}} + \frac{8\gamma}{D_i}$$

1. Isothermal Mass Conservation

Total air mass is conserved: $m_1 + m_2 = m_3$.

Since $T = \text{const}$ and $P_i V_i = m_i R T$:

$$P_1 V_1 + P_2 V_2 = P_3 V_3$$

2. Volume & Pressure Substitution

Spherical volume $V_i = \frac{\pi}{6} D_i^3$:

$$\sum_{i=1}^2 \left(p_{\text{atm}} + \frac{8\gamma}{D_i} \right) \frac{\pi}{6} D_i^3 = \left(p_{\text{atm}} + \frac{8\gamma}{D_3} \right) \frac{\pi}{6} D_3^3$$

$$\sum_{i=1}^2 (p_{\text{atm}} D_i^3 + 8\gamma D_i^2) = p_{\text{atm}} D_3^3 + 8\gamma D_3^2$$

Final Coalescence Relation

Expanding terms:

$$\begin{aligned} p_{\text{atm}} D_3^3 + 8\gamma D_3^2 &= (p_{\text{atm}} D_1^3 + 8\gamma D_1^2) \\ &\quad + (p_{\text{atm}} D_2^3 + 8\gamma D_2^2) \end{aligned}$$

Questions & Next Session: Fluid Statics